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LEARNER GUIDE

For

Build a Multi AI Agents Workflow for Stock Trading (C177)

Course Code: C177

Conducted by

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DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
2.0	25 October 2026	New course: Build a Multi AI Agents Workflow for Stock Trading - eight agents (lead, finance research, news sentiment, trading strategy, signal generation, stock trader, performance analysis, strategy review), built with AI vibe coding over eight labs; 9:30am - 5:30pm, 7.5 hours.	Dr Alfred Ang

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Introduction

This course teaches you to build a multi AI agents workflow for stock trading from scratch with AI vibe coding. You design eight cooperating agents - a lead agent that orchestrates finance research, news sentiment, trading strategy, signal generation, stock trading, performance analysis and strategy review - and you direct an AI coding assistant such as Claude Code or OpenAI Codex to write each one. You review the code against a written contract and prove it with tests before moving on.

The finished project gathers Twelve Data candles at 1-minute, 15-minute and 1-hour intervals, Yahoo Finance and Equibles company data, and Reddit and NewsAPI sentiment; formulates a strategy from trend, momentum, volatility, VWAP, RSI and MACD; places long or short orders on an Alpaca paper account; reports its performance on a dashboard; and improves its own strategy through a review loop. You finish by pushing your project to your own GitHub repository. Everything runs offline on sample data too, so no lab is blocked by a missing key. All trading is paper trading; nothing here is investment advice.

Course Learning Outcomes

- LO1: Design a multi AI agents trading workflow: the eight agent roles, the message contract between them and a lead agent that orchestrates the flow.
- LO2: Vibe-code the research agents that gather Twelve Data candles, Yahoo Finance and Equibles company data, and Reddit and NewsAPI sentiment.
- LO3: Vibe-code the decision and execution agents that formulate a strategy from trend, momentum, volatility, VWAP, RSI and MACD, generate long, short or null signals and place orders on an Alpaca paper account.
- LO4: Close the loop with a performance dashboard and a strategy review agent, orchestrate the full workflow and publish the project to GitHub.

Before You Start — Preparation

What you need

- Prerequisite: complete AI Powered Quantitative Trading Fundamentals (C728) first - it demonstrates the multi AI agents trading system and the indicators, backtesting and paper trading this course builds on.
- A laptop (Windows, macOS or Linux) with Python 3.11 or later, Git and a code editor such as Visual Studio Code.
- An AI coding assistant: Claude Code or OpenAI Codex (either works; prompts are written for both).
- A free GitHub account.
- Optional free API keys for the live labs: Twelve Data, Equibles, NewsAPI.org, a Reddit 'script' app, and an Alpaca paper-trading account. Without them every lab runs on the bundled sample data.

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- Optional: an Anthropic or OpenAI API key so the agents use an LLM brain; without it each agent uses its rule-based brain.

Verify your setup

Confirm your tools before Lab 1.

```
$ python --version     # 3.11 or later
$ git --version
$ claude --version     # or: codex --version
```

Conventions used in every lab

- Text in a grey box is either a prompt to paste into your AI coding assistant or a command to run in the terminal - the step says which.
- <YOUR-VALUE> placeholders are replaced with your own values. API keys go only in .env, never in code, prompts or screenshots.
- OFFLINE=true runs every agent on the bundled sample data and the simulated paper broker.

Topic 01 — Designing a Multi AI Agents Trading Workflow

Agent roles · message contract · lead orchestration · the AI vibe coding loop

Key concepts

- Agentic workflowSeveral specialised agents, each with one job, its own instructions and its own data tools, coordinated by a lead agent.
- The eight agentsLead, Finance Research, News Sentiment, Trading Strategy, Signal Generation, Stock Trader, Performance Analysis and Strategy Review.
- Message contractEvery agent returns one typed JSON object; the next agent reads only that object. The contract is what makes AI-written code fit together.
- AI vibe coding loopSpec -> prompt -> generate -> review -> test -> commit. You own the design and the checks; the coding agent writes the code.
- Code computes, the LLM judgesIndicators, maths and order sizes are computed in code; the LLM chooses, weighs and explains. Every agent has a rule-based fallback.
- Paper trading onlyAll orders go to an Alpaca PAPER account or the offline simulator. Nothing in this course is investment advice.

Key Concepts for Lab 1

- One role per agent: Each agent has one job, its own prompt and its own data tools - easier to prompt, test and replace.
- Typed contract first: Write messages.py before any agent: FinanceReport, SentimentReport, Strategy, TradeSignal, OrderResult, Performance, Review.
- LLM with a fallback: ask_json(system, payload, fallback, keys) returns the LLM's JSON, or the rule brain's output on any error - the workflow never stops.

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- Secrets stay in .env: Keys load from .env via python-dotenv; .env is git-ignored and never pasted into a prompt.

Lab 1 — Scaffold the Project and the Agent Message Contract

Learning outcome: LO1: design the eight-agent workflow, its message contract and the LLM helper every agent uses.

Goal: Create your own project from an empty folder with your AI coding assistant: the package layout, the .env settings, the typed message contract that every agent returns, and one ask_json() helper that calls Claude or OpenAI and falls back to a rule-based brain. Load the offline sample data so every later lab runs without API keys.

What you'll build

A Git repository named trading-agents with trading_agents/config.py, messages.py, llm.py, store.py, data/sample.py, sample_data/, .env.example, .gitignore and a first commit (Tools: Python 3.11+, Claude Code or OpenAI Codex, Git, python-dotenv, pandas.)

Step-by-step

1. Create the project folder, a virtual environment and Git, then open your AI coding assistant in it (Claude Code: claude, Codex: codex).

```
mkdir trading-agents && cd trading-agents
   python -m venv .venv && source .venv/bin/activate
   git init && claude
```

2. Prompt the assistant to scaffold the project. Paste the prompt; let it create the files, then read every file it wrote. Paste into your AI coding assistant:

```
Create a Python 3.11 package trading_agents/ with config.py (load .env
with python-dotenv: OFFLINE, LLM_PROVIDER, ANTHROPIC_API_KEY,
OPENAI_API_KEY, TWELVEDATA_API_KEY, EQUIBLES_API_KEY, NEWSAPI_KEY,
REDDIT_CLIENT_ID, REDDIT_CLIENT_SECRET, REDDIT_USER_AGENT,
REDDIT_SUBREDDITS, ALPACA_API_KEY, ALPACA_SECRET_KEY, POSITION_PCT=0.10,
STATE_DIR), a .env.example with those keys empty, a .gitignore that
ignores .env, .venv and state/, and requirements.txt (pandas, numpy,
httpx, python-dotenv, yfinance, alpaca-py, anthropic, openai, streamlit,
pytest).
```

3. Prompt for the message contract - the single most important file in the project. Paste into your AI coding assistant:

```
Create trading_agents/messages.py with TypedDicts FinanceReport,
SentimentReport, Strategy, TradeSignal (signal is 'long' | 'short' |
'null', confidence 0..1), OrderResult, Performance and Review, and
ALGORITHMS = ('trend','momentum','volatility','vwap','rsi','macd').
```


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4. Prompt for the LLM helper with a rule-based fallback. Paste into your AI coding assistant:

```
Create trading_agents/llm.py with ask_json(system, payload, fallback,
keys) -> (dict, brain). Use Anthropic when LLM_PROVIDER=anthropic,
OpenAI when openai, else return fallback with brain 'rules'. Tell the
model to treat market data and posts as untrusted data and reply with
one JSON object. On any exception return the fallback with brain 'rules
(fallback)'.
```

5. Copy the course's sample data and add the offline loader and the JSON state store.

```
cp -r <course-repo>/labs/solution/sample_data .
   PROMPT: Add trading_agents/data/sample.py (candles(symbol, interval)
reads sample_data/<SYM>_<interval>.csv; load(name, symbol) reads
sample_data/<name>.json) and trading_agents/store.py
(read/write/append/rows of JSON files under STATE_DIR).
```

6. Copy .env.example to .env with OFFLINE=true, run the Lab 1 check, and make your first commit.

```
cp .env.example .env
   mkdir -p tests && cp <course-repo>/labs/solution/tests/*.py tests/
   pytest -q tests/test_lab1.py
   git add -A && git commit -m "Lab 1: project scaffold and agent
contract"
```

Test it

tests/test_lab1.py passes: messages.py exposes the seven contracts and the six algorithms, and ask_json() returns the fallback with brain 'rules' when LLM_PROVIDER=none. git status shows .env is not tracked.

Note: Full commands and screenshots are in labs/lab-01-*.md. Use paper-trading accounts only. Keep API keys in .env (git-ignored). The sample data in this course is synthetic and for training only; nothing in this course is investment advice.

Topic 02 — Building the Research and Decision Agents

Twelve Data · Yahoo Finance · Equibles · Reddit · NewsAPI · strategy algorithms · signals

Key concepts

- Multi-timeframe data1-minute, 15-minute and 1-hour candles answer different questions: entry timing, intraday trend and session context.
- Fundamentals and alternative dataYahoo Finance company ratios and Equibles valuation, short-interest and insider data put the price action in context.

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- Sentiment as a signalReddit chatter and news headlines are scored from -1 to +1; news is weighted above social posts.
- Algorithm mixTrend, momentum, volatility breakout, VWAP, RSI and MACD each vote +1, 0 or -1; the strategy agent weights the votes and the sentiment.
- Long, short or nullA signal fires only when the combined score clears the entry threshold; otherwise the agent stays out.

Key Concepts for Lab 2

- Three intervals: 1min for entry timing, 15min for the intraday trend the indicators use, 1h for session context.
- Twelve Data: GET https://api.twelvedata.com/time_series with symbol, interval and an 'apikey' Authorization header.
- Equibles: Bearer-key REST API: /stocks/{symbol}/valuation-multiples, /short-interest and /insider-transactions.
- Indicators in code: EMA 20/50, 10-bar momentum, RSI 14, MACD 12/26/9, VWAP and Bollinger bands - computed with pandas, never by the LLM.

Lab 2 — Build the Finance Research Agent

Learning outcome: LO2: vibe-code an agent that gathers Twelve Data candles, Yahoo Finance and Equibles company data.

Goal: Direct your assistant to build three data clients - Twelve Data (1-minute, 15-minute and 1-hour candles), Yahoo Finance (company profile and ratios through yfinance) and Equibles (valuation multiples, short interest and insider transactions) - plus the indicator module, then the Finance Research Agent that combines them into one FinanceReport.

What you'll build

trading_agents/data/twelvedata.py, yahoo.py, equibles.py, trading_agents/indicators.py and agents/finance_research.py (Tools: Twelve Data API, yfinance, Equibles API, pandas, Claude Code or Codex.)

Step-by-step

1. Prompt for the Twelve Data client. Paste into your AI coding assistant:

```
Create trading_agents/data/twelvedata.py with
INTERVALS=('1min','15min','1h') and candles(symbol, interval) that calls
https://api.twelvedata.com/time_series (outputsize 300, order asc,
header Authorization: apikey <TWELVEDATA_API_KEY>) and returns an OHLCV
DataFrame indexed by timestamp. When OFFLINE=true or the key is empty,
return sample.candles().
```


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2. Prompt for the Yahoo Finance and Equibles clients. Paste into your AI coding assistant:

```
Create data/yahoo.py company(symbol) returning shortName, sector,
marketCap, trailingPE, forwardPE, profitMargins, revenueGrowth,
debtToEquity, beta and 52-week range from yfinance, and data/equibles.py
company(symbol) calling
https://api.equibles.com/v1/stocks/{symbol}/valuation-multiples, /short-
interest?limit=3 and /insider-transactions?limit=10 with a Bearer key,
recording a section's error instead of failing. Both use sample.load()
offline.
```

3. Prompt for the indicators module. Paste into your AI coding assistant:

```
Create trading_agents/indicators.py with ema, rsi, macd, vwap,
bollinger, snapshot(df) returning the latest close, ema_fast(20),
ema_slow(50), momentum_pct(10), rsi, macd, macd_signal, vwap, bb_low,
bb_high, and votes(snapshot) returning +1/0/-1 for trend, momentum,
volatility, vwap, rsi, macd.
```

4. Prompt for the agent itself. Paste into your AI coding assistant:

```
Create agents/finance_research.py with NAME='finance_research' and
run(symbol) -> (FinanceReport, brain): fetch all three intervals,
compute indicators.snapshot on the 15min candles, add Yahoo and Equibles
data, build a one-line summary as the fallback and let ask_json()
rewrite only the summary.
```

5. Review the code, run the check offline, then live with your keys, and commit.

```
pytest -q tests/test_lab2.py
   OFFLINE=false python -c "from trading_agents.agents import
finance_research as f; print(f.run('NVDA')[0]['summary'])"
   git add -A && git commit -m "Lab 2: finance research agent"
```

Test it

tests/test_lab2.py passes: the report has candles for 1min, 15min and 1h, a positive last_price, an RSI value, company and Equibles data. With keys set, the live summary names the real last price.

Note: Full commands and screenshots are in labs/lab-02-*.md. Use paper-trading accounts only. Keep API keys in .env (git-ignored). The sample data in this course is synthetic and for training only; nothing in this course is investment advice.

Key Concepts for Lab 3

- Reddit OAuth: POST https://www.reddit.com/api/v1/access_token with grant_type=client_credentials and your app's id/secret; then call https://oauth.reddit.com with 'bearer <token>' and a descriptive User-Agent.
- Where traders talk: Search r/stocks, r/wallstreetbets and r/investing for the symbol, newest first, past week.

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- NewsAPI: GET https://newsapi.org/v2/everything?q=<symbol> with the X-Api-Key header.
- Blend, then refine: Lexicon scores give a deterministic base (news 60%, Reddit 40%); the LLM may refine the score but never leaves -1..1.

Lab 3 — Build the News Sentiment Agent with Reddit and NewsAPI

Learning outcome: LO2: vibe-code the Reddit API integration and an agent that scores Reddit and news sentiment.

Goal: Integrate the Reddit API with the OAuth application-only flow, add a NewsAPI.org client and a small finance lexicon scorer, then build the News Sentiment Agent that blends both sources into one score from -1 to +1 and lets the LLM refine it.

What you'll build

trading_agents/data/reddit.py, data/newsapi.py, trading_agents/sentiment.py and agents/news_sentiment.py (Tools: Reddit API (script app), NewsAPI.org, httpx, Claude Code or Codex.)

Step-by-step

1. Create a Reddit 'script' app at reddit.com/prefs/apps and put the id, secret and a User-Agent in .env.

```
REDDIT_CLIENT_ID=<YOUR-ID>
  REDDIT_CLIENT_SECRET=<YOUR-SECRET>
  REDDIT_USER_AGENT=trading-agents/1.0 by <your-username>
```

2. Prompt for the Reddit integration. Paste into your AI coding assistant:

```
Create trading_agents/data/reddit.py: token() gets an application-only
OAuth token (client_credentials) from
https://www.reddit.com/api/v1/access_token; posts(symbol) searches each
subreddit in REDDIT_SUBREDDITS via
https://oauth.reddit.com/r/<sub>/search (q=symbol, restrict_sr=1,
sort=new, t=week) and returns subreddit, title, text, score, comments,
url. Always send REDDIT_USER_AGENT. Offline or without keys use
sample.load('reddit').
```

3. Prompt for the NewsAPI client and the lexicon scorer. Paste into your AI coding assistant:

```
Create data/newsapi.py headlines(symbol) using
https://newsapi.org/v2/everything (X-Api-Key header) and
trading_agents/sentiment.py with score(text) from small
positive/negative finance word lists and average(texts).
```

4. Prompt for the agent. Paste into your AI coding assistant:

```
Create agents/news_sentiment.py, NAME='news_sentiment', run(symbol) ->
(SentimentReport, brain): score Reddit and news separately, combine
0.4*reddit + 0.6*news as the fallback score, ask_json() may refine score
and summary, then clamp score to -1..1.
```


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5. Run the check offline and live, and commit.

```
pytest -q tests/test_lab3.py
   OFFLINE=false python -c "from trading_agents.data import reddit;
print(len(reddit.posts('NVDA')))"
   git add -A && git commit -m "Lab 3: news sentiment agent with Reddit"
```

Test it

tests/test_lab3.py passes: score is between -1 and 1 and both Reddit posts and news articles were read. With keys set, reddit.posts('NVDA') returns recent posts from the configured subreddits.

Note: Full commands and screenshots are in labs/lab-03-*.md. Use paper-trading accounts only. Keep API keys in .env (git-ignored). The sample data in this course is synthetic and for training only; nothing in this course is investment advice.

Key Concepts for Lab 4

- Six algorithms: Trend (EMA 20 vs 50), momentum (10-bar change), volatility (Bollinger breakout), VWAP (price vs VWAP), RSI (30/70 reversal) and MACD (line vs signal).
- Single or mix: The agent may pick one algorithm or weight several; weights are normalised to sum to 1.
- Combined score: (1 - sentiment weight) x weighted votes + sentiment weight x sentiment score.
- Guidance in, strategy out: DEFAULT_GUIDANCE (trend 0.4, MACD 0.35, RSI 0.25, threshold 0.3, sentiment weight 0.3) until the review agent writes guidance.json.

Lab 4 — Build the Trading Strategy Agent

Learning outcome: LO3: vibe-code an agent that merges finance and sentiment into a weighted mix of trading algorithms.

Goal: Build the agent that turns the two research reports into a strategy: the six algorithm votes come from code, the agent chooses one algorithm or a weighted mix, adds the sentiment, and reads the strategy review's guidance so the next lab's feedback loop has something to change.

What you'll build

agents/trading_strategy.py with DEFAULT_GUIDANCE and a combined score from -1 to +1 (Tools: indicators.py votes, ask_json(), Claude Code or Codex.)

Step-by-step

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1. Prompt for the agent. Paste into your AI coding assistant:

```
Create agents/trading_strategy.py, NAME='trading_strategy',
DEFAULT_GUIDANCE as described, and run(finance, sentiment, guidance) ->
(Strategy, brain). Compute votes with indicators.votes() - never ask the
LLM for indicator values. Let ask_json() choose algorithms and weights,
keep only names in ALGORITHMS, normalise weights to 1, and compute
combined_score and entry_threshold from the guidance.
```

2. Review: confirm the LLM can only choose weights, never invent votes or prices.

```
grep -n "votes" trading_agents/agents/trading_strategy.py
```

3. Try three mixes by hand and compare the combined score.

```
python -c "from trading_agents.agents import finance_research as f,
news_sentiment as n, trading_strategy as t; fi=f.run('AAPL')[0];
se=n.run('AAPL')[0]; [print(w, t.run(fi, se, {'weights': w})[0]
['combined_score']) for w in ({'trend':1}, {'rsi':1},
{'macd':0.5,'vwap':0.5})]"
```

4. Run the check and commit.

```
pytest -q tests/test_lab4.py
   git add -A && git commit -m "Lab 4: trading strategy agent"
```

Test it

tests/test_lab4.py passes: votes cover all six algorithms and the strategy's weights sum to 1. Changing the weights changes combined_score.

Note: Full commands and screenshots are in labs/lab-04-*.md. Use paper-trading accounts only. Keep API keys in .env (git-ignored). The sample data in this course is synthetic and for training only; nothing in this course is investment advice.

Key Concepts for Lab 5

- Threshold rule: long if score >= threshold, short if score <= -threshold, otherwise null.
- Confidence: The size of the combined score, clamped to 0..1.
- Guard the output: Any answer other than long, short or null is replaced by the rule signal.

Lab 5 — Build the Signal Generation Agent

Learning outcome: LO3: vibe-code an agent that turns the strategy into a long, short or null signal.

Goal: Build the small but critical agent that converts the combined score into exactly one signal with a confidence, using the entry threshold from the guidance, and guard against any invalid LLM answer.

What you'll build

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agents/signal_generation.py with rule_signal() and run() (Tools: ask_json(), Claude Code or Codex.)

Step-by-step

1. Prompt for the agent. Paste into your AI coding assistant:

```
Create agents/signal_generation.py, NAME='signal_generation',
rule_signal(score, threshold) and run(strategy) -> (TradeSignal, brain)
as specified; validate the LLM's signal and clamp confidence to 0..1.
```

2. Run the check and print the signals for three symbols.

```
pytest -q tests/test_lab5.py
```

```
   python -c "from trading_agents.agents import signal_generation as s;
print(s.rule_signal(0.5,0.3), s.rule_signal(-0.5,0.3),
s.rule_signal(0.1,0.3))"
```

3. Commit.

```
git add -A && git commit -m "Lab 5: signal generation agent"
```

Test it

tests/test_lab5.py passes and the three sample scores print long, short and null.

Note: Full commands and screenshots are in labs/lab-05-*.md. Use paper-trading accounts only. Keep API keys in .env (git-ignored). The sample data in this course is synthetic and for training only; nothing in this course is investment advice.

Topic 03 — Execution, Performance and the Feedback Loop

Alpaca paper orders · performance metrics · strategy review · lead orchestration · GitHub

Key concepts

- Automated paper executionThe stock trader agent turns a signal into a market order on Alpaca paper: long buys, short sells short, null does nothing.
- Performance metricsTotal return, Sharpe ratio, maximum drawdown and win rate measure whether the agents add value.
- Feedback loopThe strategy review agent reads the performance and rewrites the guidance the strategy agent uses next run.
- OrchestrationThe lead agent plans the run, runs the two research agents in parallel and the rest in sequence, and writes the summary.
- Publish safelyPush to GitHub with .env ignored, keys in environment variables only, a README and passing tests.

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Key Concepts for Lab 6

- Paper only: TradingClient(key, secret, paper=True) - the code never touches a live account.
- Long, short, flip: long -> buy; short -> sell short; an opposite position is closed first; already in the same direction -> skip.
- Position sizing: Shares = equity x POSITION_PCT / price, rounded down.
- Same interface offline: PaperSim keeps cash, positions and fills in state/ so the agent behaves identically without keys.

Lab 6 — Build the Stock Trader Agent on Alpaca Paper

Learning outcome: LO3: vibe-code an agent that places long and short market orders on an Alpaca paper account.

Goal: Build the broker layer - Alpaca paper trading through alpaca-py, with an offline simulated paper broker of the same interface - and the Stock Trader Agent that sizes the position and buys, sells short, flips or stays out according to the signal.

What you'll build

trading_agents/broker.py (AlpacaPaper, PaperSim, get_broker) and agents/stock_trader.py (Tools: Alpaca paper trading API (alpaca-py), Claude Code or Codex.)

Step-by-step

1. Create a free Alpaca paper account, copy the paper keys into .env.

```
ALPACA_API_KEY=<PAPER-KEY>
   ALPACA_SECRET_KEY=<PAPER-SECRET>
```

2. Prompt for the broker layer. Paste into your AI coding assistant:

```
Create trading_agents/broker.py with PaperSim (cash 100000, positions
with avg price, mark(), position(), positions(), equity(),
submit(symbol, side, qty, price) returning status and realized P&L) and
AlpacaPaper using alpaca-py TradingClient(paper=True) with the same
methods (MarketOrderRequest, TimeInForce.DAY); get_broker() returns
AlpacaPaper only when OFFLINE=false and both keys are set.
```

3. Prompt for the agent. Paste into your AI coding assistant:

```
Create agents/stock_trader.py, NAME='stock_trader', run(signal, price,
broker=None) -> (OrderResult, 'code'): null -> no order; same direction
already held -> skip; otherwise close the opposite position, then buy
(long) or sell short (short) equity*POSITION_PCT//price shares, append
each fill to trades.jsonl, and record any broker error instead of
raising.
```

4. Run the check offline, then place one live paper order and find it in the Alpaca dashboard.

```
pytest -q tests/test_lab6.py
```


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```
  OFFLINE=false python -c "from trading_agents.agents import
stock_trader as t; print(t.run({'symbol':'SPY','signal':'long'}, 580.0)
[0])"
```

5. Commit.

```
git add -A && git commit -m "Lab 6: stock trader agent on Alpaca paper"
```

Test it

tests/test_lab6.py passes: long opens a long position, short closes it and opens a short, null places no order. The live paper order appears under Orders in the Alpaca paper dashboard.

Note: Full commands and screenshots are in labs/lab-06-*.md. Use paper-trading accounts only. Keep API keys in .env (git-ignored). The sample data in this course is synthetic and for training only; nothing in this course is investment advice.

Key Concepts for Lab 7

- Total return: Latest equity / starting equity - 1.
- Sharpe ratio: Mean of per-run returns / their standard deviation x sqrt(252).
- Maximum drawdown: The worst fall from a running equity peak.
- Win rate: Share of closing trades with positive realised P&L.

Lab 7 — Build the Performance Analysis Agent and the Dashboard

Learning outcome: LO4: vibe-code an agent that computes return and risk metrics and a dashboard that shows them.

Goal: Build the agent that records the equity after every run and computes total return, Sharpe ratio, maximum drawdown, win rate and trade count, then a Streamlit dashboard that runs the workflow and shows the metrics, the equity curve, the signals and the orders.

What you'll build

agents/performance_analysis.py (metrics() and run()) and dashboard.py (Tools: Streamlit, pandas, Claude Code or Codex.)

Step-by-step

1. Prompt for the agent. Paste into your AI coding assistant:

```
Create agents/performance_analysis.py, NAME='performance_analysis',
metrics(curve, trades) returning total_return_pct, sharpe,
max_drawdown_pct, win_rate_pct and trades, and run(broker=None) that
appends the equity to equity.jsonl, computes the metrics from the equity
curve and trades.jsonl, adds open positions, lets ask_json() write a
two-sentence summary, and saves performance.json.
```


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2. Prompt for the dashboard. Paste into your AI coding assistant:

```
Create dashboard.py in Streamlit: a symbols box and a 'Run the workflow'
button, a five-metric row (equity, return, Sharpe, max drawdown, win
rate) from performance.json, a line chart of equity.jsonl, and the
strategy guidance from guidance.json.
```

3. Run the check and open the dashboard.

```
pytest -q tests/test_lab7.py
   streamlit run dashboard.py
```

4. Commit.

```
git add -A && git commit -m "Lab 7: performance analysis agent and
dashboard"
```

Test it

tests/test_lab7.py passes: a rising equity curve gives a positive return and zero drawdown; a fall gives a negative drawdown. The dashboard shows the five metrics after a run.

Note: Full commands and screenshots are in labs/lab-07-*.md. Use paper-trading accounts only. Keep API keys in .env (git-ignored). The sample data in this course is synthetic and for training only; nothing in this course is investment advice.

Key Concepts for Lab 8

- Review rules: With two or more trades and a negative return, raise the entry threshold by 0.05 and adjust; otherwise keep. The LLM may re-weight the algorithms.
- guidance.json: The review writes it; the strategy agent reads it on the next run - that is the feedback loop.
- Lead orchestration: Plan -> [finance research || news sentiment] -> strategy -> signal -> trader per symbol -> performance -> review -> summary.
- Trace everything: Every agent call is one trace entry: agent, brain (LLM or rules) and output.
- Publish: A README with the flow diagram and set-up, .env.example but never .env, tests passing, then git push.

Lab 8 — Close the Loop: Strategy Review, Lead Agent and GitHub

Learning outcome: LO4: vibe-code the strategy review feedback loop and the lead agent, then publish the project to GitHub.

Goal: Build the Strategy Review Agent that reads the performance and rewrites the strategy guidance, and the Lead Agent that orchestrates the whole workflow - the two research agents in parallel, then strategy, signal, trade, performance and review - and summarises the run. Run it twice to watch the loop change the strategy, then push your project to your own GitHub repository.

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What you'll build

agents/strategy_review.py, agents/lead.py, run.py, a README and a public GitHub repository (Tools: ThreadPoolExecutor, Git, GitHub, Claude Code or Codex.)

Step-by-step

1. Prompt for the review agent. Paste into your AI coding assistant:

```
Create agents/strategy_review.py, NAME='strategy_review',
run(performance, strategies) -> (Review, brain): start from
DEFAULT_GUIDANCE merged with guidance.json; if trades >= 2 and total
return < 0 set verdict 'adjust' and raise entry_threshold by 0.05 (max
0.6); let ask_json() suggest weights; normalise weights; save
guidance.json.
```

2. Prompt for the lead agent and the CLI. Paste into your AI coding assistant:

```
Create agents/lead.py, NAME='lead', run(symbols) that records a plan
trace entry, then for each symbol runs finance_research and
news_sentiment in parallel with ThreadPoolExecutor, then
trading_strategy (with guidance.json), signal_generation and
stock_trader; after all symbols runs performance_analysis and
strategy_review; ends with a summary from ask_json() and appends the run
to runs.jsonl. Add run.py that prints each trace step and the summary.
```

3. Run the whole workflow twice and compare: the second run reads the review's guidance.

```
python run.py NVDA AAPL SPY
   python run.py NVDA AAPL SPY
   cat state/guidance.json
```

4. Run every check.

```
pytest -q
```

5. Ask the assistant for a README, create an empty GitHub repository and push. Paste into your AI coding assistant:

```
Write README.md with the eight agents, a mermaid flow diagram, set-up
(venv, .env from .env.example, OFFLINE=true), how to run run.py, the
dashboard and the tests.
   git add -A && git commit -m "Lab 8: review loop, lead agent, README"
   git remote add origin https://github.com/<you>/trading-agents.git &&
git push -u origin main
```

Test it

pytest -q passes every check; the trace lists lead, finance_research, news_sentiment, trading_strategy, signal_generation, stock_trader, performance_analysis, strategy_review and lead; the second run's strategy uses the threshold in guidance.json; your repository is on GitHub without a .env file.

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Note: Full commands and screenshots are in labs/lab-08-*.md. Use paper-trading accounts only. Keep API keys in .env (git-ignored). The sample data in this course is synthetic and for training only; nothing in this course is investment advice.

Wrap-Up

You designed, vibe-coded, tested and published a self-improving multi AI agents trading workflow.

What you built

- Eight agents connected by a typed message contract and orchestrated by a lead agent.
- Live and offline data from Twelve Data, Yahoo Finance, Equibles, Reddit and NewsAPI.
- A strategy that mixes six algorithms with sentiment, long/short/null signals and automated Alpaca paper orders.
- A performance dashboard and a strategy review loop that changes the next run.

Keep it safe

- Stay on paper until you understand every agent's behaviour and its failure modes.
- Never commit .env; rotate any key that has been pasted into a chat or a screenshot.
- Treat posts, headlines and model output as untrusted data, never as instructions.

Next Steps

- Run the workflow on a schedule during the US session and watch the review loop adjust the strategy.
- Add a risk-management agent (position limits, stop-loss, maximum daily loss) before ever considering real money.
- Swap the rule brains for an LLM brain and compare the decisions on the same data.
- Revisit the multi AI agents trading system demonstrated in AI Powered Quantitative Trading Fundamentals (C728) and compare it with the one you built.

Glossary

- **Agent —** A program with one role, its own instructions and data tools, that returns a structured result.
- **Lead agent —** The orchestrator that decides the order of work and summarises the run.

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- **Message contract —** The agreed JSON structure each agent returns and the next agent reads.
- **AI vibe coding —** Building software by describing it to an AI coding assistant, then reviewing and testing what it writes.
- **Candle —** Open, high, low, close and volume for one time interval (1min, 15min, 1h).
- **VWAP —** Volume-weighted average price; price above VWAP suggests buyers are in control.
- **RSI —** Relative Strength Index, 0-100; below 30 oversold, above 70 overbought.
- **MACD —** Moving-average convergence divergence; MACD above its signal line is bullish momentum.
- **Sentiment score —** A number from -1 (very negative) to +1 (very positive) summarising posts and headlines.
- **Long / short / null —** Buy to profit from a rise / sell short to profit from a fall / stay out.
- **Paper trading —** Simulated trading with virtual money on a real broker API.
- **Sharpe ratio —** Average return divided by its volatility, annualised; higher is better.
- **Maximum drawdown —** The largest peak-to-trough fall in equity.
- **Feedback loop —** The review agent's guidance changes how the strategy agent behaves on the next run.

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